

Question 1 — Regular expressions (10 marks)

(a) Consider the following text:

```
Order #4471 shipped on 2026-03-14; order #4502 on 2026-04-02.
```

1. Write a regular expression that captures the year, month and day of each date as three separate groups. (2 marks)
2. The pattern `\d+` is used to extract the order numbers. List everything it matches in this text, explain why, and correct the pattern so that it returns only `4471` and `4502`. (3 marks)

(b) Consider the following text:

```
The theatre is near the other gate.
```

1. Count the matches of the pattern `the` in this text, and name each word in which a match occurs. (2 marks)
2. Write a pattern that matches only the word *the* in either case, and explain what each part of your pattern contributes. (3 marks)

Question 2 — Tokenisation (10 marks)

(a) Consider the following sentence:

Mr. Das can't pay \$12.50 today.

1. Tokenise the sentence by splitting only at spaces, and state the number of tokens. (2 marks)
2. Write the tokens a Penn Treebank-style rule-based tokeniser should produce. Explain how it treats `Mr.`, `can't` and `$12.50`, and why. (3 marks)

(b) The number is the word's frequency; `_` marks the end of a word.

```
6  s i t _  
3  s i t s _  
2  f i t _
```

1. Perform **two** BPE merges, stating the pair and its count for each. Then apply the merges in order to segment the unseen word `f i t s _`. (3 marks)
2. Explain why BPE never needs an `<UNK>` token for a word made of known characters, and state one cost of performing too few merges. (2 marks)

Question 3 — Normalisation, stemming and lemmatisation (10 marks)

(a) A pipeline lower-cases its input and removes the stop words {a, the, of, us}.

The Head of IT emailed US staff about a new policy

1. Apply the pipeline and write the tokens that remain. (2 marks)
2. Identify two pieces of information this pipeline destroys, and explain for each whether a search application would be harmed. (3 marks)

(b) Consider the tokens:

running ran runs runner

1. Write the output of a Porter-style suffix-stripping stemmer and of a lemmatiser (treating every token as a verb where possible) for each token. (3 marks)
2. Using your answer, explain which of the two groups *ran* with *running*, and why the other cannot. (2 marks)

Question 4 — Part-of-speech tagging and named entities (10 marks)

(a) Consider the following sentence:

She quickly signed the forms .

1. Assign a Penn Treebank POS tag to every token. (3 marks)
2. Give the POS tag of *run* in They run daily and in It was a long run , and explain what information a tagger must use to tell them apart. (2 marks)

(b) Consider the following sentence:

Infosys opened an office in Mysuru last March .

1. Tag every token with the BIO scheme, using the types ORG, GPE and DATE. (3 marks)
2. Explain why the BIO scheme needs a separate B tag rather than marking every entity token with I , using two adjacent entities of the same type as your example. (2 marks)

Question 5 — The term-document incidence matrix (10 marks)

Use this collection for both parts:

D1	wheat	rice	price	D4	rice	price	ban
D2	rice	export		D5	wheat	price	
D3	wheat	export	ban				

(a)

1. Write the incidence vectors for `wheat`, `rice` and `price`. (2 marks)
2. Use bitwise operations on the vectors to answer `wheat AND price`, and `(wheat OR rice) AND NOT ban`. Show each step. (3 marks)

(b)

1. A collection has 1,000,000 documents and a vocabulary of 500,000 terms. Compute the number of cells in its incidence matrix. (2 marks)
2. Each document contains about 1,000 distinct terms. Estimate the fraction of cells that hold a 1, and explain what this implies about how the matrix should be stored. (3 marks)

Question 6 — The inverted index and TF-IDF (10 marks)

(a) Use the collection of Question 5.

1. Write the postings lists, with document frequencies, for `rice`, `price` and `ban`. (2 marks)
2. Answer `rice AND price AND ban` by merging postings. State the order in which you process the three terms, and explain why that order is chosen. (3 marks)

(b) A collection has $N = 1,000,000$ documents. Weights are computed with $\text{tf-idf} = (1 + \log_{10} \text{tf}) \times \log_{10}(N/\text{df})$.

term	tf in document	df
monsoon	10	1,000
report	1,000	100,000

1. Compute the tf-idf weight of each term in the document. (3 marks)
2. `report` occurs 100 times as often as `monsoon` in the document. Explain why it still receives the smaller weight, and what the logarithm on tf contributes. (2 marks)

Question 7 — N-gram probabilities (10 marks)

A bigram model is trained on this corpus, which is also used in Question 8:

```
<s> i want tea </s>  
<s> i want food </s>  
<s> you want tea </s>  
<s> i like food </s>
```

(a)

1. Write $P(\text{you want food})$ as a product of bigram probabilities, including both boundary symbols, and evaluate it. (3 marks)
2. Evaluate $P(\text{i want tea})$ in the same way, and explain why it is larger than $P(\text{you want food})$. (2 marks)

(b)

1. Estimate $P(\text{food} \mid \text{<s> you want})$ using the full history, and compare it with the bigram estimate $P(\text{food} \mid \text{want})$. (2 marks)
2. Explain what the comparison shows about the Markov assumption, and why n-gram models make it despite its cost. (3 marks)

Question 8 — The zero problem and smoothing (10 marks)

Use the corpus of Question 7.

(a)

1. Compute $P(\text{you like tea})$ with the unsmoothed bigram model. (2 marks)
2. Identify every bigram responsible for your answer, and explain why this result is a problem for evaluating the model on a test set. (3 marks)

(b)

1. Recompute $P(\text{like} \mid \text{you})$ with add-one smoothing, stating V and what it counts. (2 marks)
2. Compute $P(\text{want} \mid \text{you})$ before and after add-one smoothing, and explain what the change shows about how add-one smoothing redistributes probability. (3 marks)

Question 9 — Interpolation and backoff (10 marks)

(a) For the word *coffee*, a model has these estimates:

$$\begin{aligned} P(\text{coffee} \mid \text{she drinks}) &= 0 & P(\text{coffee} \mid \text{drinks}) &= 0.4 \\ P(\text{coffee}) &= 0.05 \end{aligned}$$

1. Compute the interpolated probability with $\lambda_1 = 0.5$, $\lambda_2 = 0.4$, $\lambda_3 = 0.1$.
(2 marks)
2. Give the estimate the simplified backoff procedure taught in class would use (ignoring discounting), and compare the two answers. (3 marks)

(b)

1. A student proposes $\lambda_1 = 0.6$, $\lambda_2 = 0.3$, $\lambda_3 = 0.2$. Explain what is wrong with these weights and what would happen to the model's probabilities.
(2 marks)
2. Explain how the λ values should be chosen, and why choosing them on the training corpus would push λ_1 towards 1. (3 marks)

Question 10 — Perplexity (10 marks)

(a) A bigram model assigns the test sentence `<s> i want tea </s>` the probability $1/1296$.

1. State N for this sentence and compute the perplexity. (3 marks)
2. A model that treats all 10 tokens of its vocabulary as equally likely is scored on the same sentence. Compute its perplexity and explain what the comparison tells us. (2 marks)

(b) Two models are scored on the same test set of $N = 3$ tokens. Model A assigns it the probability $1/64$ and Model B the probability $1/27$.

1. Compute the perplexity of each model and state which is better. (3 marks)
2. A third model assigns probability 0 to one bigram in the test set. State its perplexity, and explain why a model with this property is not used in practice. (2 marks)

End of model questions.